

Week 2 Discussion: Quiz 1 Review & Practice (SOLUTION KEY)

Topics: Option definitions, Payoff vs. Profit calculations, Long/Short positions, and Leverage.

1 Rapid Review: The Equations You Need

1. What is an Option?

An **Option** is a financial contract that gives the buyer (holder) the **right**, but not the obligation, to buy or sell an underlying asset at a specific price (Strike Price K) on or before a specific date (Expiration Date T).

- **Call Option:** The right to **BUY**.
- **Put Option:** The right to **SELL**.

2. Position Summary (Long vs. Short)

Position	View	Payoff Formula	Risk Profile
Long Call	Bullish ($\nearrow$)	$\max(S_T - K, 0)$	Limited Loss, Unlimited Gain
Short Call	Bearish ($\searrow$)	$-\max(S_T - K, 0)$	Unlimited Loss , Limited Gain
Long Put	Bearish ($\searrow$)	$\max(K - S_T, 0)$	Limited Loss, Significant Gain
Short Put	Bullish ($\nearrow$)	$-\max(K - S_T, 0)$	Significant Loss, Limited Gain
Short Forward	Bearish ($\searrow$)	$K - S_T$	Unlimited Loss (Linear)

3. Profit vs. Payoff (The Most Common Mistake)

- **Payoff:** The cash flow from the contract **at expiration**. (Does NOT include initial cost).
- **Profit:** The net money you make.

$$\text{Profit} = \text{Payoff} - \text{Initial Cost (Premium)}$$

- *Note: For Forward contracts, Initial Cost is usually 0, so Profit = Payoff.*

2 Concept: Leverage (Why trade options?)

Options allow you to control more shares with less capital, magnifying both gains and losses.

Problem 1: Stock vs. Option

You have **\$1,000** to invest. You expect TSLA stock ($S_0 = \$100$) to rise.

- **Strategy A (Stock):** Buy 10 shares of TSLA.
- **Strategy B (Option):** Buy Call Options ($K = 100$) for \$5 each. You can buy 200 options.

Scenario 1: TSLA rises to \$110 (10% increase).

(a) Calculate the ROI for Strategy A.

$$\text{Profit} = (110 - 100) \times 10 = \$100. \quad \text{ROI} = \frac{100}{1000} = 10\%$$

(b) Calculate the ROI for Strategy B. (Payoff per option = $110 - 100 = \$10$).

$$\text{Profit} = (10 - 5) \times 200 = \$1,000. \quad \text{ROI} = \frac{1000}{1000} = 100\%$$

Conclusion: A 10% move in stock price caused a 100% return in options.

Scenario 2: TSLA falls to \$99 (1% decrease).

(a) Strategy A Loss: $(99 - 100) \times 10 = -\10 (Loss of 1%).

(b) Strategy B Loss: Options expire worthless. Loss = **\$1,000** (Loss of 100%).

3 Quiz Practice Problems

Problem 3: Comprehensive Option Calculation

Fill in the table below. Assume you enter the following positions:

- **Position 1:** Long Call ($K = 50$, Cost = \$4)
- **Position 2:** Short Put ($K = 50$, Premium Received = \$3)

Spot Price (S_T)	Long Call ($K = 50, c = 4$)		Short Put ($K = 50, p = 3$)	
	Payoff	Net Profit	Payoff	Net Profit
\$40	0	$0 - 4 = -4$	$-(50 - 40) = -10$	$-10 + 3 = -7$
\$48	0	-4	-2	1
\$50	0	-4	0	3
\$55	5	1	0	3

Questions:

(a) For the **Long Call**, at what stock price do you break even? ($S_T = ?$)

Break-even Price = Strike + Premium = $50 + 4 = \$54$.

(b) For the **Short Put**, what is the **maximum possible profit** you can make?

Max Profit = Premium Received = \$3 (Happens if $S_T \geq 50$).

(c) True or False: In the Short Put position, you have the *right* to buy the stock.

False. The Short position has the *obligation* to buy if the Long side exercises.

Problem 4: Conceptual Check

Explain why a "Short Call" position is considered riskier than a "Long Put" position, even though both are bearish strategies (betting on price going down).

Solution: A Long Put has limited risk; the most you can lose is the premium paid. A Short Call has theoretically unlimited risk because there is no upper limit to how high a stock price can rise ($S_T \rightarrow \infty$), meaning the payoff you owe ($-\max(S_T - K, 0)$) can be infinitely large.